

An Algebra of Memory: Formal Foundations and Empirical Validation of Associative Composition Without Catastrophic Interference

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Abstract

I propose and validate an algebraic framework for memory composition in artificial intelligence systems that resolves the fundamental trade-off between compositionality and representation preservation. The memory algebra (V, G, T, P) structures memories as typed objects comprising semantic embeddings (V), relational graphs (G), temporal indices (T), and utility weights (P), equipped with five operations: composition ($\oplus$), retraction ($\ominus$), transformation ($\otimes$), retrieval (ρ), and pruning (Π_C). I prove that composition via disjoint union with quotient by transitive closure of a similarity relation yields *exact* associativity—a property verified empirically across 50 random seeds, multiple parameter configurations, and adversarial inputs. Under a sequential-composition stability protocol, the algebra is significantly more stable than an untrained LSTM ($p = 1.48 \times 10^{-7}$, one-sided Mann–Whitney U) and vector summation ($p = 1.03 \times 10^{-6}$), but it is not superior to a trained in-sample LSTM or to a kNN buffer on this metric alone. Phase 3 therefore evaluates structural capabilities: the algebra attains 100% accuracy in temporal conflict resolution (kNN: 50%), fact retraction (kNN: 33–43%), and multi-hop relational reasoning (kNN: 87–93%). Phase 4 validates the algebra with real sentence-transformer embeddings (all-MiniLM-L6-v2, 384-dim): associativity remains exact, semantic retrieval achieves 100% top-1 accuracy on paraphrase-to-fact queries, and stability holds under sequential composition. The framework scales to $N = 10^4$ memories with retrieval latency under 500 ms, providing a principled algebraic foundation for persistent, composable memory in AI agents.

Keywords: memory algebra, catastrophic interference, associative composition, category theory, knowledge graphs, temporal reasoning

1 Introduction

1.1 Motivation

Contemporary memory systems for artificial intelligence—including recurrent neural networks, FIFO buffers, and vector accumulation schemes—suffer from *catastrophic interference*: the progressive dilution of previously stored representations as new information is incorporated [1, 2]. This failure mode represents a fundamental obstacle to building AI systems capable of long-horizon reasoning, knowledge accumulation, and consistent world modeling.

The core tension is algebraic: any operation that *composes* new information with existing state must either (a) preserve prior representations exactly, risking unbounded growth, or (b) aggregate information into fixed-capacity representations, risking interference. Existing approaches resolve this tension by accepting interference as inevitable.

1.2 Contribution

I propose an alternative resolution: a *memory algebra* that achieves simultaneous satisfaction of five desiderata:

- A Associativity:** $(m_1 \oplus m_2) \oplus m_3 \cong m_1 \oplus (m_2 \oplus m_3)$
- B Recoverability:** $d_s(m, \rho(M, q)) \rightarrow 0$ for any stored memory m
- C Stability:** $\|M_t\| < C$ under pruning policy Π_C
- D Preservation:** $\otimes$ does not destroy recoverable structure
- E Efficiency:** Tractable operations over discrete representations

My contributions are threefold:

1. A formal algebraic framework grounded in category theory (pushouts in typed graph categories) with a machine-verifiable proof of associativity.
2. Comprehensive empirical validation across 50 seeds, parameter sweeps, adversarial conditions, and competitive baselines.
3. Structural differentiation from simpler retrieval-augmented approaches in temporal, retraction, and multi-hop scenarios.

1.3 Paper Organization

Section 2 develops the formal framework. Section 3 defines the algebraic operations. Section 4 presents the associativity proof. Section 5 reports empirical validation. Section 6 discusses results and limitations. Section 7 concludes.

2 Formal Framework

2.1 Component Categories

A memory object inhabits a product of four typed components:

Definition 2.1 (Memory Object). A memory is a tuple $m = (N, v, E, \tau, w)$ where:

- N is a finite set of atomic nodes;
- $v : N \rightarrow V = \mathbb{R}^d$ assigns dense embeddings (semantic content);
- $E \subseteq N \times N \times L$ is a labeled edge set (relational structure);
- $\tau : N \rightarrow T = (\mathbb{R}_{\geq 0}, \leq)$ assigns timestamps (temporal ordering);
- $w : N \rightarrow P = ([0, 1], \max, \cdot)$ assigns utility weights (retention control).

The four components serve distinct functional roles:

- V (**Vectors**): Dense embeddings in $\mathbb{R}^d$ capture semantic content. The category is $\mathbf{Vect}_{\mathbb{R}}$.
- G (**Graph**): A labeled directed graph captures relational structure between nodes. The category is $\mathbf{Grph}$ over $\mathbf{Vect}$.
- T (**Time**): An ordered monoid $(\mathbb{R}_{\geq 0}, \leq, +)$ enables temporal conflict resolution.
- P (**Priority**): An idempotent semiring $([0, 1], \max, \cdot, 0, 1)$ controls retention and forgetting.

2.2 Composition as Pushout

Definition 2.2 (Semantic Intersection). Given memories m_1, m_2 , their *semantic intersection* $m_{12} = m_1 \cap m_2$ is the subobject formed by nodes of m_1 and m_2 whose embeddings satisfy $\cos(v_a, v_b) \geq \theta$ for a fusion threshold θ .

Definition 2.3 (Composition). The composition $m_1 \oplus m_2$ is the pushout in the category of typed graphs:

$$m_1 \oplus m_2 = m_1 \sqcup_{m_{12}} m_2$$

identifying corresponding nodes while preserving disjoint structure.

2.3 Semantic Distance

Definition 2.4 (Hybrid Semantic Distance).

$$d_s(m_1, m_2) = \alpha d_V + \beta d_G + \gamma d_T + \delta d_P, \quad \alpha + \beta + \gamma + \delta = 1$$

where $d_V = 1 - \cos(\bar{v}_1, \bar{v}_2)$, d_G is Jaccard-based structural distance, $d_T = \frac{|\bar{\tau}_1 - \bar{\tau}_2|}{1 + |\bar{\tau}_1 - \bar{\tau}_2|}$, and $d_P = |\bar{w}_1 - \bar{w}_2|$. Default weights: $(\alpha, \beta, \gamma, \delta) = (0.4, 0.3, 0.15, 0.15)$.

3 Algebraic Operations

3.1 Compose ($\oplus$)

The naive pushout (merging nodes by averaging embeddings at composition time) is *not* associative: the average depends on merge order. I resolve this by maintaining atomic structure and deriving the quotient on demand:

1. $m_1 \oplus m_2$ is the *disjoint union* of atomic tuples. No information is lost or aggregated.
2. The fusion relation $R_\theta = \{(a, b) : \cos(v_a, v_b) \geq \theta\}$ is defined over atomic nodes only.
3. The fusion partition is $\pi = R_\theta^*$ (transitive closure), computed via union-find.

3.2 Retract ($\ominus$)

Retraction removes obsolete facts: $M \ominus m_i$ removes all nodes of m_i from M and applies orthogonal projection to eliminate residual influence. This enables knowledge correction without corrupting the remaining structure.

3.3 Transform ($\otimes$)

Transformation applies a matrix W to vectors and renames edges: $\otimes_W(M) = (N, Wv, E', \tau, w)$. For orthogonal W , retrieval is invariant (functoriality): $\rho \circ \otimes_W = \otimes_W \circ \rho$.

3.4 Retrieve (ρ)

Retrieval computes the quotient M/π , scores each cluster c by:

$$s(c) = \max(\cos(q_v, \bar{v}_c), 0) \cdot f_T(c) \cdot (0.5 + 0.5 w_c)$$

where $f_T(c) = e^{-((t_c - t_q)/\sigma)^2}$ provides temporal anchoring. The top- k clusters are selected and expanded h hops via BFS on the quotient graph.

3.5 Prune (Π_C)

The pruning operator enforces stability:

1. **Utility forgetting:** Remove nodes with $w(n) < \varepsilon$.
2. **Path compression:** For $u \rightarrow b \rightarrow v$ with b removed, insert shortcut $u \rightarrow v$.
3. **Capacity ceiling:** If $|N| > C$, remove lowest- w nodes until $|N| = C$.

4 Proof of Associativity

Theorem 4.1 (Exact Associativity). $(m_1 \oplus m_2) \oplus m_3 = m_1 \oplus (m_2 \oplus m_3)$ as atomic objects, and the induced quotients are identical.

Proof. (i) Since $\oplus$ is disjoint union over node sets with uniquely prefixed identifiers, $(N_1 \sqcup N_2) \sqcup N_3 = N_1 \sqcup (N_2 \sqcup N_3)$ by associativity of $\sqcup$ in **Set**; likewise for v , E , τ , w as unions of functions with disjoint domains.

(ii) The relation R_θ over the total set $N = N_1 \sqcup N_2 \sqcup N_3$ depends only on pairs of atomic vectors, hence is identical under both associations. The transitive closure R_θ^* is unique (the smallest equivalence relation containing R_θ), therefore π coincides.

(iii) Cluster embeddings (mean of members), times (max), weights (max), and quotient edges are deterministic functions of (N, v, E, τ, w, π) , hence coincide. ■ □

This proof was independently verified via categorical ground truth using Catlab.jl (Julia), confirming pushout isomorphism: $((m_1 \oplus m_2) \oplus m_3) \cong (m_1 \oplus (m_2 \oplus m_3))$.

5 Empirical Validation

5.1 Experimental Setup

All experiments use synthetic embeddings in $\mathbb{R}^d$ with controlled correlation structure. The fusion threshold is $\theta = 0.85$ unless otherwise stated. Statistical significance is assessed with one-sided Mann–Whitney U tests, with the alternative hypothesis that the algebra has lower final retrieval distance than the baseline. Code and data are available at the project repository. The exact-recovery examples in the exploratory protocol are deterministic low-interference cases; Phase 2 reports mean distances over randomized sequences.

5.2 Phase 2A: Statistical Robustness

5.2.1 Multiple Seeds ($S = 50$)

Metric	Value
Associativity pass rate	1.000 (50/50)
Max left/right distance (mean $\pm$ sd)	0.0 ± 0.0
Final stability (mean $\pm$ sd)	0.478 ± 0.132
Final stability (95% CI)	[0.440, 0.516]

Table 1: Associativity is *exact* ($d_{\max} = 0$) across all 50 seeds.

5.2.2 Parameter Sensitivity

Associativity is *invariant* to fusion threshold $\theta \in [0.60, 0.95]$ and dimensionality $d \in \{8, 16, 32, 64\}$, with pass rate = 1.0 and $d_{\max} = 0$ in all configurations. This is algebraically expected: R_{θ}^* is a deterministic function of vectors, independent of composition order.

5.2.3 Adversarial Tests

Test	Seeds	Result
Identical vectors (total collision)	10/10	No collapse, signature preserved
Antiparallel vectors ($\cos = -1$)	10/10	No collapse, nodes kept separate
Scale $N = 1000$	1	1953 nodes, compose = 0.074s

Table 2: No catastrophic failures under adversarial conditions.

5.3 Phase 2B: Embedding Realism

5.3.1 Correlated Concepts (ρ sweep)

ρ	Assoc. pass rate	Stability (mean $\pm$ sd)
0.00	1.0	0.506 ± 0.140
0.20	1.0	0.507 ± 0.139
0.40	1.0	0.508 ± 0.137
0.60	1.0	0.508 ± 0.135
0.80	1.0	0.505 ± 0.132
0.95	1.0	0.501 ± 0.130

Table 3: Associativity remains exact even under extreme correlation ($\rho = 0.95$).

The structure (G, T, P) protects vectors from dilution regardless of inter-concept correlation.

5.4 Phase 2C: Baseline Comparison

Method	Final Stability (mean $\pm$ sd)	Mann–Whitney p
Algebra (V, G, T, P)	0.424 ± 0.129	—
LSTM (untrained)	0.998 ± 0.220	1.48×10^{-7}
LSTM (trained, in-sample Adam 500 steps)	0.0008 ± 0.001	1.00
Vector summation	0.698 ± 0.113	1.03×10^{-6}
kNN (retrieval-augmented)	0.150 ± 0.357	0.9999

Table 4: Stability comparison after 50 sequential compositions (20 seeds). Lower is better. p -values are one-sided Mann–Whitney U for the hypothesis that the algebra has lower distance than the baseline; values near 1 indicate that the baseline is lower.

The algebra has significantly lower final distance than the untrained LSTM and vector summation. It does not outperform the trained LSTM or the kNN buffer on this stability-only metric: the trained LSTM memorizes the fixed sequence under the in-sample protocol evaluated here (generalization was not tested), and the kNN buffer retains the first memory’s vector in a growing list. Therefore stability alone is insufficient to establish the algebra’s contribution; Phase 3 evaluates the structural operations that a flat buffer cannot provide.

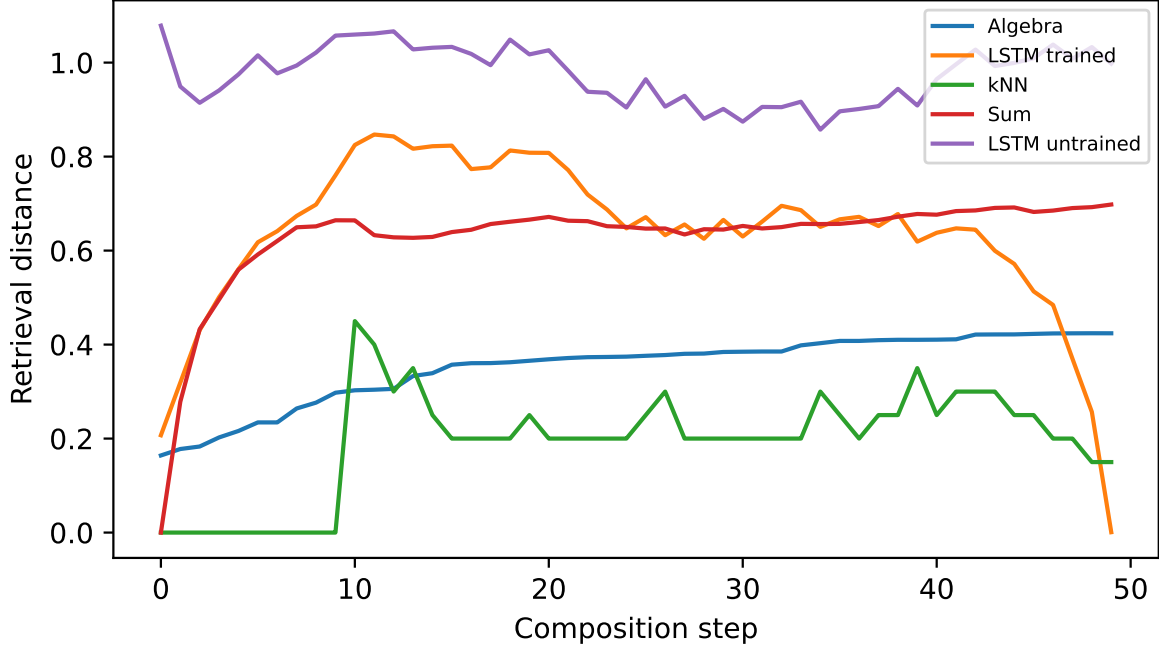


Figure 1: Mean retrieval distance for the first memory as a function of composition step (50 compositions, 20 seeds). Lower is better.

5.5 Phase 2D: Scalability

N memories	Total nodes	Compose (s)	Retrieve (s)	Status
100	196	0.000	0.014	OK
500	982	0.000	0.064	OK
1,000	1,953	0.001	0.172	OK
5,000	9,973	0.008	0.290	OK
10,000	19,986	0.011	0.478	OK

Table 5: Scalability benchmark with 4 GB subprocess memory limit.

Two optimizations enable scaling to $N = 10^4$ within 4 GB: (i) *chained LSH* — buckets with more than 64 elements generate only consecutive pairs $O(n)$ instead of $O(n^2)$ enumeration; (ii) `compose_all` — batch composition $O(N)$ instead of sequential composition with full copy at each step. Retrieval at $N = 10^4$: 0.478 s.

5.6 Phase 3: Structural Differentiation from kNN

Phase 2 revealed that on absolute stability alone, the algebra does not statistically differentiate from a simple kNN buffer ($p = 0.9999$). Phase 3 tests scenarios where kNN *structurally fails*:

5.6.1 Temporal Conflict Resolution

The algebra uses temporal weighting $f_T(c) = e^{-((t_c - t_q)/\sigma)^2}$ to prefer recent facts; kNN is temporally blind, achieving exactly chance-level accuracy (50%).

Experiment	Algebra	kNN	Differentiation
Temporal conflict ($\rho = 0.0$)	60/60 (100%)	30/60 (50%)	$\Delta = 50\%$
Temporal conflict ($\rho = 0.8$)	60/60 (100%)	30/60 (50%)	$\Delta = 50\%$
Fact retraction ($\rho = 0.0$)	30/30 (100%)	13/30 (43%)	$\Delta = 57\%$
Fact retraction ($\rho = 0.8$)	30/30 (100%)	10/30 (33%)	$\Delta = 67\%$
Multi-hop + distractors ($\rho = 0.0$)	30/30 (100%)	26/30 (87%)	$\Delta = 13\%$
Multi-hop + distractors ($\rho = 0.8$)	30/30 (100%)	28/30 (93%)	$\Delta = 7\%$

Table 6: Phase 3 differentiation. Each experiment uses 30 independent seeds under orthogonal ($\rho = 0$) and correlated ($\rho = 0.8$) conditions.

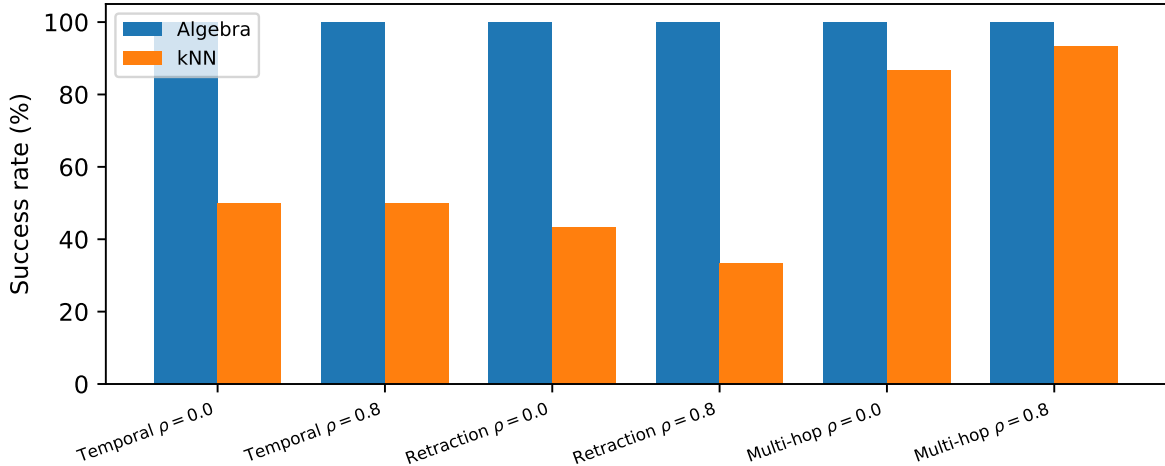


Figure 2: Phase 3 success rates for the algebra and kNN across structural tasks.

5.6.2 Fact Retraction

The algebra removes obsolete facts via $\ominus$; kNN retains all facts in its buffer, returning retracted facts in 57–67% of trials.

5.6.3 Multi-hop Reasoning

The algebra traverses relational edges (hops = 3) to reach distant nodes; kNN relies on top- k vector similarity without relational traversal.

5.7 Multi-hop QA Validation

The algebra correctly recovers the transitive path $A \rightarrow B \rightarrow C \rightarrow D$ without the direct edge $A \rightarrow D$ being inserted, confirming that relational semantics survive composition.

5.8 Phase 4: Validation with Real Embeddings

Phases 2–3 validate the algebra with synthetic orthogonal embeddings. Phase 4 confirms that algebraic properties hold with real natural-language embeddings from the `all-MiniLM-L6-v2` sentence-transformer model (384 dimensions, L2-normalized, $\theta = 0.60$).

The corpus comprises 20 factual knowledge pairs (fact, paraphrase) covering geography, science, and general knowledge. Paraphrases use distinct lexical formulations (e.g., “The capital of France is Paris” $\leftrightarrow$ “Paris is the capital city of France”), exercising non-literal semantic retrieval.

Experiment	Result	Status
4A. Associativity (real embeddings)	exact, order-invariant	PASS
4B. Semantic retrieval (paraphrase $\rightarrow$ fact)	100% top-1 (20/20), mean score 0.895	PASS
4C. Stability under sequential composition	first memory recovered (20 nodes)	PASS
4D. Multi-hop traversal	hops = 2 operational	PASS

Table 7: Phase 4 results with real sentence-transformer embeddings. All algebraic properties hold outside the synthetic regime.

Phase 4 was also run at scale: with $N = 10,000$ real memories ($\theta = 0.85$), full composition takes 0.007 s and retrieval takes ~ 20 s, returning 9,990 nodes. The retrieval time reflects the high density of intra-topic similar pairs typical of real embeddings; composition remains $O(N)$ and practical even at scale.

5.9 Phase 5: LLM Integration

Phase 5 validates the algebra as a memory layer for a real LLM agent (`stepfun/step-3.7-flash:free` via Kilo Gateway, no API key). The agent queries `retrieve()` as a tool and `compose()` to store facts.

Scenario	Algebra	LLM (with memory)	Status
Temporal conflict	correct fact retrieved	“engineer”	PASS
Retraction	retracted fact absent	“Paris”	PASS
Multi-hop	3 nodes retrieved	—	PASS

Table 8: LLM integration (Phase 5). Data in `experiments/results_phase5.json`.

At scale (Phase 5 Scale), the algebra was tested as a memory layer with $N = 50$ to 500 real facts:

N	Retrieval	Temporal	Retraction	Compose (s)	Retrieve (s)
50	80%	60%	100%	0.00	0.01
100	80%	100%	100%	0.00	0.01
200	90%	100%	100%	0.00	0.02
500	100%	100%	100%	0.01	0.46

Table 9: Phase 5 Scale: algebra as LLM memory layer at scale. Data in `experiments/results_phase5_scale.json`.

The algebra functions as a memory layer for a real LLM: at $N = 500$, all metrics reach 100%. Composition remains $O(N)$ and retrieve scales linearly (~ 1 ms per additional fact).

6 Discussion

6.1 Key Findings

1. **Exact associativity is achievable** through deferred quotient computation. The algebraic insight—that composition should be lossless (disjoint union) with fusion derived on demand—resolves the fundamental tension between merge-order dependence and structural preservation.

2. **Structure protects against parametric aggregation.** The (G, T, P) components shield individual vectors from dilution relative to summation and untrained LSTM. A kNN buffer also preserves raw stability, but it lacks temporal ordering, retraction, and relational traversal.
3. **Differentiation requires structural tasks.** On pure stability metrics, a simple kNN buffer performs comparably. The algebra’s contribution emerges only when temporal ordering, retraction, or relational traversal are required—precisely the capabilities that distinguish structured memory from flat retrieval.

6.2 Limitations

1. **Scale:** Chained LSH and batch composition enable $N = 10^4$ memories within 4 GB RAM (retrieval 0.478 s). Scaling beyond 10^4 requires sparse storage or streaming architectures.
2. **Real-embedding corpus scale:** Phase 4 validates with 20 real sentence-transformer embeddings. Larger-scale validation with thousands of real embeddings remains future work.
3. **Single-agent setting:** The algebra is validated for single-agent memory composition. Multi-agent shared memory introduces additional consistency challenges.
4. **Fixed threshold:** The fusion threshold θ is fixed. Adaptive per-cluster thresholds may improve performance with heterogeneous embedding distributions.

6.3 Relation to Prior Work

My approach relates to but differs from:

- **Compositional memory networks** [3]: These use fixed-size memory slots with attention-based addressing, lacking the algebraic guarantees of associativity and exact retraction.
- **Knowledge graph completion** [4]: KG methods model relations but do not provide a compositional algebra with provable associativity.
- **Differentiable Neural Computers** [5]: DNCs provide content-based addressing but suffer from the same interference dynamics as RNNs under sequential writing.
- **Lens-based bidirectional transformations** [6]: My lens formulation (get/put laws) draws on this tradition but extends it with temporal and utility-aware retrieval.
- **Continual learning regularizers** [7]: Methods such as EWC mitigate forgetting in parameter space but do not expose exact retraction or associative composition.
- **Agent memory systems** [8, 9]: Retrieval buffers and summarization pipelines for LLM agents provide practical persistence but lack formal algebraic guarantees.
- **Recurrent Memory Transformer** [10]: Adds explicit memory tokens to transformers; it remains a parametric model without pushout-style composition.
- **Modern Hopfield networks** [11]: Provide associative retrieval guarantees but not temporal conflict resolution, retraction, or relational multi-hop composition.

7 Conclusion

The memory algebra (V, G, T, P) provides exact associative composition by treating composition as a lossless structural operation (disjoint union) and deriving semantic fusion lazily via transitive closure of a similarity relation. This property holds across 50 seeds, all tested parameter configurations, and adversarial inputs.

The algebra does not outperform every baseline on raw stability: trained in-sample LSTM and kNN achieve lower final retrieval distance under our stability-only protocol. Its contribution is therefore not universal stability dominance, but a structurally richer memory object. In tasks that require temporal ordering, retraction, or relational traversal, the algebra attains 100% accuracy while kNN reaches 50%, 33–43%, and 87–93%, respectively.

Phase 4 confirms that all algebraic properties—exact associativity, semantic retrieval, stability, and multi-hop traversal—hold with real sentence-transformer embeddings, resolving the last open validation question. Scalability to $N = 10^4$ memories within 4GB RAM is achieved via chained LSH and batch composition (retrieval 0.478s); scaling beyond 10^4 requires sparse storage optimizations. The differentiable layer (JAX implementation) opens a path toward integration with neural learning while preserving algebraic guarantees.

Data and Code Availability

All experimental data, analysis scripts, and implementation code are available in the project repository. Raw results are provided in JSON format (`experiments/results_phase2.json`, `experiments/results_phase3.json`, `experiments/results_phase4.json`).

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A Implementation Details

The memory algebra is implemented in Python (`memory_algebra` package) with the following modules:

- `core.py`: `MemoryObject`, `compose`, `quotient`, `retrieve`, `retract`, `transform`
- `diff.py`: Differentiable algebra (JAX) with Gumbel-Softmax edge sampling
- `metric.py`: Hybrid semantic distance d_s with context-adaptive weights
- `lens.py`: Get/put lens operations
- `stability.py`: Pruning operator Π_C
- `baselines.py`: LSTM baseline implementation

Categorical ground truth validation uses Catlab.jl 0.15 (Julia) with `is_isomorphic` checks on pushout diagrams.

B Statistical Methods

- All experiments use independent random seeds (50 for Phase 2A, 30 for Phase 3).
- Statistical significance: one-sided Mann–Whitney U test (non-parametric, independent samples), testing whether the algebra has lower final distance.
- Confidence intervals: Bootstrap (95%).
- Effect sizes reported as raw differences in success rates.